YISSUM האוניברסיטה העברית בירושלים THE HEBREW UNIVERSITY OF JERUSALEM

Open Dashboard →

HUJI Applicative Research Report — Agriculture & Food — Week 45, 2026

1 paper(s) · Agriculture & Food · Weekly · Week 45, 2026 · Generated November 02, 2026

■■ There is no high-potential applicable research this period — only lower-scoring papers were found. Presenting the two best below.

This week's highlight focuses on groundbreaking AgriTech advancements from Hebrew University, specifically targeting precision agriculture. New research offers a refined understanding of plant respiration, a fundamental process impacting growth and resource use. This foundational science is poised to enable the development of high-precision sensors and predictive AI for optimizing crop yields and enhancing carbon sequestration strategies. Such innovations are critical for addressing global food security and climate challenges.

RESEARCHERS & RESEARCH SUBJECTS — BEST AVAILABLE

- **Yotam Zait** — AgriTech, Software/AI

AGRICULTURE & FOOD — RESEARCH HIGHLIGHTS

1. Estimating daytime respiration with steady-state and dynamic Laisk methods and its implications for mesophyll conductance. 14/50

"Precision plant respiration modeling will power next-gen AgriTech for yield optimization and carbon sequestration."

Main researcher: Yotam Zait · yotam.zait@mail.huji.ac.il

The Robert H. Smith Institute of Plant Sciences and Genetics in Agriculture, Faculty of Agriculture, Food and Environment, The Hebrew University of Jerusalem, Rehovot 76100, Israel.

Published in Tree physiology · 2026 Oct 30

ABOUT THIS RESEARCH

This research focuses on improving the precision of measuring how plants breathe (respiration) during the day using advanced mathematical modeling. By refining these measurements, researchers can better understand how gases move within plant tissues and how they respond to environmental changes.

COMMERCIAL ANGLE

The ability to more accurately model plant gas exchange can be leveraged to develop high-precision sensors and predictive software for optimizing crop yields and carbon sequestration in precision agriculture.

COMMERCIALISATION METRICS

Novelty 3/10

The paper focuses on applying and comparing established physiological methodologies (Laisk methods) rather than introducing a fundamentally new mechanism or a paradigm-shifting technology. It represents incremental refinement and comparative analysis within existing plant physiology frameworks.

Commercial Potential 3/10

This is a fundamental physiological study focused on methodology and biophysical parameters rather than a developed product or scalable technology. While useful for agricultural modeling, it lacks a clear pathway to a proprietary device, service, or licensable platform.

Market Size 3/10

The paper describes a niche methodological advancement in plant physiology research rather than a scalable commercial product or platform. The addressable market is limited to academic and specialized botanical research institutions with low industrial demand.

Tech Readiness 2/10

The paper focuses on fundamental physiological modeling and methodology refinement for plant respiration, which is at the basic research/proof-of-concept stage. It lacks the hardware integration or field-validated application required for higher TRL levels.

IP Strength 3/10

The paper describes mathematical modeling and physiological estimation methods based on established biophysical principles (Laisk methods), which are generally considered academic discovery rather than patentable novel hardware or proprietary compositions. While scientifically valuable for plant physiology, it lacks the structural or algorithmic novelty typically required for strong industrial IP defensibility.

View in Dashboard Open Paper